

MUHAMMAD TAIMOOR TARIQ

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Research Interests

My research focuses on **resource management and coordination in multi-tenant networks**, with a specific interest in designing architectures that reconcile the tension between efficiency, fairness, and performance isolation. I develop systems that enable diverse tenants with conflicting objectives to share common underlying infrastructure without degrading individual performance.

Education

University of Illinois, Urbana-Champaign

August 2022

PhD. in Computer Science

Urbana, IL

- Advised by: Radhika Mittal

Lahore University of Management Sciences (LUMS)

September 2018 – May 2022

Bachelor of Science in Computer Science

Lahore, Pakistan

- Placed on the Dean's Honors List for academic excellence for years 2020-2022

Publications

Saving Private Cellular: Addressing The Coordination Vacuum in CBRs

Taimoor Tariq, Radhika Mittal

HotMobile 2026

Performance Isolation for 5G RAN Slicing Across Multiple Interfering Cells

Taimoor Tariq, Yongzhou Chen, Haitham Hassanieh, Radhika Mittal

NINeS 2026

Centralized Traffic Engineering for Networked Farm Applications

Ammar Tahir, Yueshen Li, Jianli Jin, Changxin Zhang, Daniel Moon, Aganze Mihigo, Taimoor Tariq, Deepak Vasisht, Radhika Mittal

IEEE / ACM SEC 2025

Enabling Emerging Edge Applications Through a 5G Control Plane Intervention

Mukhtiar Ahmad, Muhammad Nawazish Ali, Taimoor Tariq, Basit Iqbal, Taqi Raza, Zafar Ayyub Qazi

ACM CoNEXT 2022

Neutrino: A Fast and Consistent Edge-based Cellular Control Plane

Mukhtiar Ahmad, Muhammad Nawazish Ali, Taimoor Tariq, Syed Usman Jafri, Adnan Abbass, Syeda Mashal Abbas, Basit Iqbal, Zartash Uzmi, Zafar Ayyub Qazi

IEEE/ACM Transactions on Networking, 2022

Fast Failure Detectors for 5G Edge Deployments

Taimoor Tariq, Maha Kamal, Vafa Batool

Student Research Competition, SOSP 2021

Industry Experience

Google

May 2025 – August 2025

PhD Software Engineering Intern – Hosts: Yingjie Bi, Hamid Bazzaz

Sunnyvale, CA

- Designed a simulation framework to estimate reservable bandwidth for Premium Bandwidth in datacenter clusters.

Teaching Experience

Teaching Assistant

August 2024 – December 2024

CS-425: Distributed Systems

UIUC, IL

Teaching Assistant

January 2022 – May 2022

CS-678: Topics in Internet Research (Graduate)

LUMS, Pakistan

Lead Teaching Assistant

September 2021 – December 2021

CS-582: Distributed Systems (Graduate)

LUMS, Pakistan

Teaching Assistant

January 2021 – May 2021

CS-382: Network-Centric Computing (Undergraduate)

LUMS, Pakistan